

MCU Pin Assignment — S32K148 to the 94-way Bosch Connector

Date: 2026-09-19 **Status:** Phase 2, first artifact. Unblocks the per-pin fail-safe detail spec §4 deferred and the two Phase-2 items memo 05 §7 left open (how many ADC/FTM channels the 144-pin package actually breaks out; per-pin reset defaults for the six gate pins).

0. Preamble — package, sources, method

Package assumed: 144-pin LQFP, orderable part `FS32K148HAT0MLQT` .

This is the package already selected in spec §5 and confirmed in `docs/research/05-mcu-selection.md` — chosen as headroom against the 50 still-unresolved OEM connector pins (U1's register), not because the *confirmed* 44-pin budget needs it. This document does not revisit that choice; it uses the `S32K148_144lqfp` column of NXP's own pin table throughout.

Documents retrieved this session, fresh, not from memory:

- NXP, *S32K1xx Data Sheet*, Rev. 15, 5 March 2026 — `nxp.com/docs/en/data-sheet/S32K1xx.pdf` , the same revision `docs/research/05-mcu-selection.md` cites. Used for the LVR/LVD electrical thresholds (§3 below) and to confirm the datasheet itself defers the pinout table to the Reference Manual (§10.1, "For package pinouts and signal descriptions, refer to the Reference Manual" — this is a direct quote, not a paraphrase, and it is why memo 05 could not close its own two Phase-2 items from the datasheet alone).
- NXP, *S32K1xx Series Reference Manual*, Rev. 13, 04/2020 — retrieved from `community.nxp.com` (the same mirror memo 05 found but did not open). **This PDF has eleven Excel workbooks embedded as file attachments**, including `S32K148_IO_Signal_Description_Input_Multiplexing.xlsx` — the actual chip-specific pin-mux table the Reference Manual's own Chapter 4 says exists but does not inline ("IO Signal Description Input Multiplexing sheet(s) attached to the Reference Manual"). Extracted with `pdfdetach` and read directly (893 rows, `IO Signal Table` tab). **This is the primary source for every `PTxx` ↔ peripheral ↔ package-pin-number claim in this document, all tagged CONFIRMED below**, and it is the single fact this whole exercise turned on: the manual's *narrative* text was retrievable by search all along, but the actual per-pin table was not, because it lives in an attachment nobody had opened.

- Sensor front-end topology and signal counts:

docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md §4/§5, refs/ecu-pinout-extracted.md , sim/blocks/*.cir (read, not modified), docs/research/05-mcu-selection.md , docs/research/12-driver-architecture.md .

One revision gap, disclosed rather than smoothed over. The Reference Manual pulled this session is Rev. 13 (04/2020); the datasheet memo 05 cites is Rev. 15 (2026). No higher-revision Reference Manual was located or checked. Pin multiplexing is a silicon mask property, not a document-editing one, and no revision-history entry in either document (datasheet's own changelog was searched) mentions a pinout or signal-mux change for the S32K148 specifically — but this is **INFERRED**, not independently confirmed against a Rev.-15-class Reference Manual, because none was found. If a newer RM surfaces before schematic capture, diff its

S32K148_I0_Signal_Description_Input_Multiplexing.xlsx against the one this document used before trusting a pin unchanged.

Package pin budget, checked rather than assumed: the embedded table lists 157 PTxx port pins across the whole S32K1xx family tree; 128 of them are bonded out on the 144-pin LQFP package (the rest exist only on the 176-pin part or the BGA). This document commits 39 of those 128 to a signal (37 as first written, plus the two watchdog pins §1.8 adds). That leaves 89 GPIO-capable pins unused — enough margin that the 50 still-unknown OEM connector pins (spec §8) will not run the package out of pins once their function is known, which was an open question memo 05 flagged and this answers: **the 144-pin package was the right headroom call.**

Reserved and excluded from this map, deliberately: PTA4 / PTC4 / PTC5 (JTAG TMS/TCK/TDI) and PTA5 (dedicated RESET_b) are the only four pins on the whole part with a reset default other than high-impedance-disabled (Reference Manual §4.4, Table 4-3 — see §2 below). None of the 94 connector signals are placed on them. PTB6 / PTB7 (XTAL / EXTAL) are not used by any signal here either, but whether this design needs an external crystal at all is not decided anywhere in the repo — flagged in §5.

1. The mapping table

Columns: ECU pin (Bosch 94-way, per spec §4 / refs/ecu-pinout-extracted.md), Signal, Front-end (from spec §4 / sim/blocks/ , not re-derived here), PTxx , Peripheral, Reset pull (CONFIRMED from the embedded IO Signal Table — every non-JTAG/RESET pin reads PE=0,

PS=0 , i.e. **disabled, high-impedance, no pull**, matching RM §4.4 Table 4-3's blanket "Others: Disabled, High impedance" row), and a **Note** giving the one-line reason for anything not a free choice.

A pin bearing "-" in the PTxx column is not an MCU signal at all — it is a power, ground, or excitation-supply net that terminates at the power tree or the sensor-ground plane, not at a GPIO. Those are listed so that every one of spec §4's 44 confirmed connector pins is accounted for, not silently dropped.

1.1 Power

ECU pin	Signal	Front-end	P Tx x	Peripheral	Reset pull	Note
21	Battery +	Reverse-battery FET → buck	—	—	—	Power input net, not an MCU signal
04	V_BAT_1R sense	120k/10k divider (spec §4)	P TA 0	ADC0_SE0	Hi-Z, no pull	Free choice; paired with pin 06 on the <i>other</i> ADC instance (below) so both battery rails sample simultaneously
06	V_BAT_2R sense	120k/10k divider	P TA 2	ADC1_SE0	Hi-Z, no pull	Paired with 04 for simultaneity — a real S32K148 property (two independent SAR ADCs), not decorative
09	SYNCHRONIZATION GROUND	—	—	—	—	U6 open. Function unconfirmed; likely a ground per its own name despite being drawn red. No MCU pin assigned — assigning one would be guessing a signal that may not exist

1.2 Analog sensors

ECU pin	Signal	Front-end	P Tx x	Peripheral	Reset pull	Note
11	Boost sensor 5V excitation	5V_SENSOR distribution	—	—	—	Power net, not an MCU input
41	Boost pressure	Differential (shared gnd 34)	P TA 1	ADC0_SE1	Hi-Z, no pull	
79	Boost temperature	Differential or NTC — U2 open, topology TBD	P TA 3	ADC1_SE1	Hi-Z, no pull	One ADC pin serves either candidate front-end in <code>ntc_frontend.cir</code> ; the pin choice does not depend on U2's answer
34	Sensor ground (boost+coolant)	Instrumentation-amp reference	—	—	—	Analog ground-plane net, not a GPIO
33	Coolant temperature	Differential (shared gnd 34)	P TA 6	ADC0_SE2	Hi-Z, no pull	
15	EGR 5V excitation	5V_SENSOR distribution	—	—	—	Power net

ECU pin	Signal	Front-end	P Tx x	Peripheral	Reset pull	Note
37	EGR position	Differential (shared gnd 36)	P TA 1 5	ADC1_SE12	Hi-Z, no pull	
36	Sensor ground (EGR+oil)	Instrumentation-amp reference	—	—	—	Analog ground-plane net
39	Oil pressure 5V excitation	5V_SENSOR distribution	—	—	—	Power net
80	Oil pressure	Differential (shared gnd 36)	P TA 7	ADC0_SE3	Hi-Z, no pull	
32	Rail pressure 5V excitation	5V_SENSOR distribution	—	—	—	Power net
35	Rail pressure	Single-ended (dedicated gnd 08)	P TA 1 6	ADC1_SE13	Hi-Z, no pull	
08	Sensor ground (rail, dedicated)	—	—	—	—	Analog ground-plane net
—	5V_SENSOR rail monitor	Divider → ADC (spec §4's ratiometric-correction channel)	P TC 1 4	ADC0_SE12	Hi-Z, no pull	No connector pin — internal PCB net, required by the firmware ratiometric-correction scheme spec §4 already specifies
—	Onboard barometric sensor	New, per spec §7 deviation #2	P TD 1 8	ADC1_SE16	Hi-Z, no pull	No connector pin — this is the 8th "sensor input" in memo 05's 14-channel ADC budget

1.3 Speed

ECU pin	Signal	Front-end	PTxx	Peripheral	Reset pull	Note
52 / 74	Crank VR High/Low	Differential adaptive-threshold conditioner (vr_conditioner.cir)	(conditioner input, no direct MCU pin)	—	—	The conditioner's zero-cross comparator already collapses this to one digital edge stream — see next row
—	Crank conditioner output	Zero-cross comparator output	PTB2	FTM1_CH0 (input capture)	Hi-Z, no pull	Forced in kind, not in pin: whatever pin is chosen must be FTM/capture-capable; PTB2 is a free choice among many valid options. FTM configured for both-edge capture, per RM's CnSC[ELSA:ELSB]
30	Crank sensor ground	Shield termination, ECU end	—	—	—	Ground/shield net
45	Cam Hall 5V excitation	5V_SENSOR distribution	—	—	—	Power net

ECU pin	Signal	Front-end	PTxx	Peripheral	Reset pull	Note
46	Cam frequency I/P	Pull-up to 5V_SENSOR per spec §4	PTB3	FTM1_CH1 (input capture)	Hi-Z, no pull	Flagged in §4 below — this front-end as literally specified overvoltages the MCU pin
44	Cam sensor ground	—	—	—	—	Ground net

1.4 Discrete inputs

ECU pin	Signal	Front-end	P Tx x	Peripheral	Reset pull	Note
43	Coolant switch	Pull-up + RC debounce	PTD5	GPIO input	Hi-Z, no pull	
23	Droop switch	Pull-up + RC debounce	PTD7	GPIO input	Hi-Z, no pull	
01	G_G_B AT1	—	—	—	—	Unresolved per spec §4's own note — memo 03 suspects this is a power ground, not a switch-return leg. No MCU pin assigned either way: if it's a ground it needs none, and if it is a signal, guessing which one would violate this project's own "nothing guessed into copper" rule
02	G_G_B AT2	—	—	—	—	Same as 01
20	Audio abort (active-high)	47k/68k bias + 3.0V zener clamp	PTD11	GPIO input	Hi-Z, no pull	
24	Override SS (active-high)	Same clamp front-end	PTD12	GPIO input	Hi-Z, no pull	
71	Ignition (active-high)	Same clamp front-end	PTD16	GPIO input	Hi-Z, no pull	
—	Trip-module sense (new)	trip_module_sense.cir — 47k/68k/220nF/3.0 V, same chain as the rows above	PTB0	ADC0_SE4	Hi-Z, no pull	CORRECTED 20 Sep 2026, during schematic capture — was PTD6 GPIO. See below

CORRECTION, 20 September 2026 — the trip sense needs an ADC, not a GPIO.

This row originally read PTD6, GPIO input, "placeholder pin only — front-end is undesigned". The front-end has since been designed, and designing it invalidated the pin class.

sim/blocks/trip_module_sense.cir builds a **three-state supervised loop** and checks all three bands with real margin:

State	Node voltage, swept 6–40 V
healthy — contact closed, bleed resistor shorted	2.863 – 2.954 V
tripped — contact open, 1.2 MΩ bleed in circuit	0.310 – 2.068 V
wire fault — nothing reaches the divider	0.000 V

A GPIO reads two states. It separates healthy from not-healthy and nothing else, which throws away exactly the distinction the bleed resistor exists to create. A supervised loop that cannot tell a trip from a cut wire is an unsupervised loop with extra parts in it.

PTD6 is not ADC-capable — checked against `refs/s32k148-144lqfp-altfn.csv`, and neither is any other pin in §1.4, which is correct for the channels there and wrong for this one. The channel moves to PTB0 (ADC0_SE4 / ADC1_SE14): free, bonded out on the 144-LQFP at pin 78, PE=0 / PS=0 at reset. PTD6 returns to the spare pool.

Committed count is unchanged at 39; the spare GPIO count is unchanged at 89.

1.5 Outputs — the six gate pins are in §2

ECU pin	Signal	Front-end	PTx x	Peripheral	Reset pull	Note
03	Injector HS bank A (cyl 1+3)	Boosted high-side switch	PTC 0	FTM0_CH0	Hi-Z, no pull	See §2
05	Injector HS bank B (cyl 2)	Boosted high-side switch	PTC 1	FTM0_CH1	Hi-Z, no pull	See §2
73	Injector LS cyl 1	Current-controlled low-side	PTC 2	FTM0_CH2	Hi-Z, no pull	See §2
07	Injector LS cyl 3	Current-controlled low-side	PTC 3	FTM0_CH3	Hi-Z, no pull	See §2
29	Injector LS cyl 2	Current-controlled low-side	PTB 4	FTM0_CH4	Hi-Z, no pull	See §2
88	Fuel metering PWM	High-current low-side + current sense	PTD 10	FTM2_CH0	Hi-Z, no pull	See §2
59	EGR High (bridge IN1)	H-bridge	PTB 8	FTM3_CH0	Hi-Z, no pull	Not one of the six — see §2's scope note
81	EGR Low (bridge IN2)	H-bridge	PTB 9	FTM3_CH1	Hi-Z, no pull	Not one of the six
50	Main relay	Low-side FET + flyback	PTD 13	GPIO output	Hi-Z, no pull	
69	Buzzer relay	Low-side FET + flyback	PTD 14	GPIO output	Hi-Z, no pull	

1.6 Current-sense channels (internal, no connector pin)

Memo 05's ADC budget names "4 current-sense returns from the injector, metering and EGR drivers." These are shunts on the driver stage, not connector signals, so they have no ECU pin — but they need an MCU PTxx and are placed here for completeness.

Associated signal	P Tx x	Peripheral	Reset pull	Note
Injector bank A current sense (pins 03/73)	P TC 1 5	ADC0_SE13	Hi-Z, no pull	INFERRED split: assumes one shunt per high-side bank, not per low-side cylinder. Memo 12 says "per channel or per bank" without deciding; this map picks per-bank because that is what makes the 4-channel figure in memo 05/\$5 add up (2 banks + metering + EGR = 4). Three-channel per-cylinder sensing would need a 5th ADC pin — trivially available, but not what the existing budget assumes. Settle by choosing the shunt placement when the injector driver sheet is drawn.
Injector bank B current sense (pin 05, low side 29)	P TD 1 9	ADC1_SE17	Hi-Z, no pull	Paired with bank A on the other ADC instance for simultaneous sampling
Metering unit current sense (pin 88)	P TC 1 6	ADC0_SE14	Hi-Z, no pull	
EGR bridge current sense (pins 59/81)	P TD 2 2	ADC1_SE18	Hi-Z, no pull	

1.7 CAN

Both channels are real (spec §2.5). The S32K148 has three FlexCAN instances; this design needs two, exactly as memo 05 notes ("1 spare instance"). CAN0/CAN1 are used, CAN2 is left spare — a free choice among three equally-valid instances, not a forced one.

ECU pin	Signal	PTxx	Peripheral	Reset pull	Note
55	CAN channel 1, H	(via transceiver)	—	—	Bus-side signal; transceiver CANH pin, not an MCU pin
77	CAN channel 1, L	(via transceiver)	—	—	Transceiver CANL pin
—	CAN channel 1, MCU side	PTE5	FlexCAN0_TX	Hi-Z, no pull	To transceiver 1 TXD
—	CAN channel 1, MCU side	PTE4	FlexCAN0_RX	Hi-Z, no pull	From transceiver 1 RXD
87	CAN channel 2, H	(via transceiver)	—	—	Transceiver 2 CANH
86	CAN channel 2, L	(via transceiver)	—	—	Transceiver 2 CANL
—	CAN channel 2, MCU side	PTC7	FlexCAN1_TX	Hi-Z, no pull	To transceiver 2 TXD

ECU pin	Signal	PTxx	Peripheral	Reset pull	Note
—	CAN channel 2, MCU side	PTC6	FlexCAN1_RX	Hi-Z, no pull	From transceiver 2 RXD
—	CAN channel 3 (spare)	(unassigned)	FlexCAN2	—	Instance exists, no pins committed

1.8 SPI / supervisor interface

No SPI-dependent part is currently decided anywhere in the repo. The handbook's speculative peripheral table (docs/handbook/sensors.html) once counted "1 SPI channel for the injector driver," but memo 12 (19 Sep 2026) decided discrete FETs with a conventional gate-driver IC for the injectors and a smart low-side switch for the metering unit — neither is a microcode/SPI-programmed part in memo 12's recommendation. So: **no SPI pin is committed in this map.** LPSPI1 (PTB14 SCK / PTB15 SIN / PTB16 SOUT / PTB17 PCS3) is entirely free and is the natural reservation if a future EEPROM, external diagnostic port, or SPI-configured driver IC is added — but forcing a pin assignment ahead of a part choice would be exactly the kind of guess this project's own discipline rules out.

The supervisor interface is not a data bus at all — that is the point of it. sim/blocks/supervisor.cir and docs/research/11-supervisor-and-fail-safe-outputs.md specify a TPS3850-class windowed supervisor whose RESET output drives the S32K148's own RESET_b pin and, through one inverting stage, a dedicated kill transistor at each of the six gate nodes — entirely in analog/discrete hardware, with no MCU involvement in that path by design (that is what lets it protect against a hung MCU).

CORRECTED 20 Sep 2026, during schematic capture. This section previously said "the only MCU pin in this interface is RESET_b ", and that was wrong in a way that mattered: the kill path needs no MCU pin, but the watchdog does. A window watchdog is not a passive monitor — the MCU has to present a falling edge on WDI inside the window, every window, or the part asserts and resets the board on a timer. A supervisor topology with no pin to kick it is a supervisor that guarantees a reset loop.

Two pins are therefore added here, both taken from the 91 spare GPIO the map leaves free, both GPIO-HD , both PE=0 / PS=0 at reset (confirmed from the same embedded IO Signal Table as every other row in this document):

Signal	P Tx x	Package pin	Peripheral	Reset pull	Note
RESET_b	PTA5	141	dedicated	weak pull-up	Forced — this is the only RESET_b pin on the part. Dedicated function, not muxed. Reset default per RM Table 4-3: weak pull-up enabled (the one deliberate exception among the pins this document touches)
WDT_KICK	PT0	138	GPIO output	Hi-Z, no pull	MCU → TPS3850 WDI . Free choice of pin; what is <i>not</i> free is that some pin must do this
WDT_FAULT	PT1	137	GPIO input	Hi-Z, no pull	TPS3850 WDO → MCU, so firmware can tell a watchdog timeout from a rail excursion. WDO asserts only while RESET is high, which is what makes the two distinguishable

That takes the committed count from 37 signals to 39, and the spare GPIO from 91 to 89.

The supervisor part is also now chosen, which §1.8 previously left as a class: **TPS3850G33** (`TPS3850G33DRCT` , VSON-10). The variant matters and the reasoning is in `hw/gen_symbols.py` beside the symbol — briefly, the datasheet's own nomenclature table gives `G` = thresholds at $\pm 4\%$ of nominal and `H` = $\pm 7\%$, and with the part's $\pm 0.8\%$ accuracy that puts G33's worst-case undervoltage trip at 3.143 V against H33's 3.044 V. The supervisor has to assert *before* the S32K148's own LVD (3.0 V maximum), so G33 clears it by 143 mV and H33 by 44 mV — close enough that the two could fire in either order, and a supervisor that might lose the race to the thing it supervises is not doing its job. The cost is a tighter window the 3V3 rail has to live inside (3.143–3.459 V), which is a constraint this map now places on the rails design.

2. The six gate pins and their reset behaviour

Spec §4's binding requirement (added 18 Sep 2026) says every driver gate on pins 88, 73/07/29, 03/05 must be held off by hardware during reset, because S32K148 GPIOs are high-impedance out of reset. The requirement explicitly left its own per-pin detail open:

"the physical PTxx assignment for pins 88, 73/07/29 and 03/05 is a phase-2 task (§5), so the per-pin reset pull defaults cannot yet be looked up."

That is now answered, pin by pin, from the manufacturer's own IO Signal Table (not inferred):

ECU pin	PTxx	PE (pull enable)	PS (pull select)	Reset state
88	PTD10	0	0	Disabled, high-impedance
73	PTC2	0	0	Disabled, high-impedance
07	PTC3	0	0	Disabled, high-impedance

ECU pin	PTxx	PE (pull enable)	PS (pull select)	Reset state
29	PTB4	0	0	Disabled, high-impedance
03	PTC0	0	0	Disabled, high-impedance
05	PTC1	0	0	Disabled, high-impedance

CONFIRMED against both the Reference Manual's own prose (§4.4, Table 4-3: "Others: Disabled — High impedance — `ibe=0`, `obe=0`, `pue=0`, `pus=0` ") and the per-pin `PE` / `PS` / `Reset` fields in the embedded `S32K148_I0_Signal_Description_Input_Multiplexing.xlsx`, read directly for each of these six pins.

None of the six defaults to a pull that works against the fail-safe requirement — and none defaults to a pull that helps it either. There is no pull at all. The only four pins on the entire part with any reset-time pull are `PTA4` / `PTC4` / `PTC5` (JTAG) and `PTA5` (`RESET_b`) — none of which are used for gate duty. So the exposure spec §4 and memo 11 already reasoned about from first principles (a floating gate, brought up by leakage or Miller coupling, with nothing internal to the MCU resisting it) is **exactly the confirmed condition for all six pins**, not a worst-case assumption that turned out pessimistic or optimistic. This closes the open item cleanly: the hardware kill-clamp `supervisor.cir` already validated does not change, because it was designed to not depend on this answer, and the answer it did not depend on turned out to match the design's own worst case precisely.

Scope note on EGR (pins 59/81). Spec §4's fail-safe list names only the six pins above — not the EGR bridge. A concurrently-updated file, `sim/blocks/egr_hbridge.cir`, argues EGR's own safe state is different in kind (coast, not off, so a spring return can close the valve) and is drafting its own gate-kill treatment. That file is outside this task's scope and was not modified here; it is noted so the two documents are not read as disagreeing by omission.

3. Datasheet numbers reconfirmed, not just cited

Spec §4 and memo 11 state the S32K148's brownout thresholds as VLVR 2.50–2.7 V, VLVD 2.8–3.0 V, and a 2.97 V PLL-engaged correctness-guarantee floor. Re-read directly from the Rev. 15 datasheet's own electrical characteristics tables this session (not taken on trust from the earlier memo):

- `VLVR` (LVR falling threshold, RUN/HSRUN/STOP): 2.50 / 2.58 / 2.7 V (min/typ/max)
- `VLVD` (falling low-voltage detect threshold): 2.8 / 2.875 / 3.0 V

- "S32K148 will operate from 2.7 V when executing from internal FIRC. When the PLL is engaged S32K148 is guaranteed to operate from 2.97 V." — quoted directly, twice, at different points in the datasheet (electrical characteristics footnote and I/O operating-voltage table, both consistent).

CONFIRMED, exact match to spec/memo 11's figures. No arithmetic correction needed here — recorded because the task asked for datasheet numbers to be checked, not carried forward on trust a second time.

4. What did not fit, and what is INFERRED

Cam signal front-end (pin 46) — a finding, not a fix. Spec §4 describes pin 46's front-end as "Pull-up to 5V_SENSOR → timer capture," i.e. the Hall sensor's open-drain output pulled directly to the 5 V sensor rail, into an MCU timer-capture pin. Read literally, that puts a 5 V logic swing on a 3.3 V-rail MCU pin — over the S32K148's absolute maximum ($V_{DD} + 0.3\text{ V}$, roughly 3.6 V at $V_{DD} = 3.3\text{ V}$, per the datasheet's own operating-voltage figures in §3 above). No `sim/blocks/*.cir` file **simulates a cam front-end at all** (the built-block list in `docs/status.md` §1.3 has `vr_conditioner.cir` for crank; there is no cam equivalent) — so this was never checked, and the pin map surfaces it rather than papering over it with a pin choice. PTB3 is assigned as the *capture pin*, but the front-end feeding it needs a level clamp or divider before the pad, matching the discipline already applied to pins 04/06 (battery sense) and 20/24/71 (discrete inputs). **This should become a `sim/blocks/cam_frontend.cir`-class check before the cam sheet is drawn, not solved here.**

Injector current-sense split (§1.6) is INFERRED, not confirmed. Memo 12 leaves "per channel or per bank" open; this map assumes per-bank (2 channels) because that is the only split consistent with memo 05's stated "4 current-sense returns" total once metering (1) and EGR (1) are subtracted. A per-cylinder scheme (3 channels) is equally buildable and would just need one more free ADC1 pin — **settle when the injector driver sheet's shunt placement is actually drawn.**

RM revision currency (§0) is INFERRED. Rev. 13 (04/2020) was the newest S32K148 IO signal table located; the datasheet elsewhere cited is Rev. 15 (2026). No pin-mux errata were found in either document's revision history, but a newer Reference Manual was not checked because none was found. Low risk (pinout is a silicon property), not zero.

Crystal reservation (PTB6 / PTB7) is an open question, not a finding. Nothing in the repo states whether this design runs off the internal FIRC or an external crystal. Neither pin is used by any signal in this map, so there is no conflict either way — but if CAN bit-timing accuracy (a real automotive concern, not evaluated here) ends up requiring a crystal reference, those two pins should be reserved before anything else claims them.

SPI (§1.8): no pin committed, for the reason stated there — no SPI-dependent part is decided. LPSP11 is free and is the natural landing spot.

5. Open questions

1. **Cam front-end overvoltage (§4)** — needs a `sim/blocks/cam_frontend.cir` -class check and a level-clamp design before pin 46's sheet is drawn. This is new, found by this exercise, not inherited from an existing unknown.
2. **Injector current-sense placement (§1.6)** — per-bank vs per-cylinder, decided when the injector driver sheet's shunt is placed.
3. **Crystal vs FIRC (§4)** — decides whether PTB6 / PTB7 are reserved.
4. **U6 (pin 09) and pins 01/02** — still open per spec §8/§4; no MCU pin assigned, consistent with "nothing guessed into copper."
5. **The 50 unconfirmed OEM connector pins (spec §8)** — cannot be placed on PTxx without a function to place. 91 of the 128 GPIO-capable pins on the 144-pin package remain uncommitted, which is comfortable headroom once they are known.
6. **RM revision currency (§0)** — worth a diff against a newer Reference Manual if one surfaces before layout.